

Louvers

Louvers are the unsung heroes of building ventilation, specialized in controlling the flow of air in and out of your structure. louvers come in various types, each with a unique function seamlessly blending with your building's exterior. Weatherproof louvers are designed to prevent rain from entering the building, while sand trap louvers stop sand from entering with the fresh air into the HVAC system, Acoustic louvers to damp unwanted noise.

Types:

- **Weatherproof louvers (FAL):** designed to prevent rain from entering the building. They are typically made of extruded aluminum (std.) or SST or galvanized steel as an option. Their blades are angled downward horizontally to deflect rain this method is called Cascade Method.
- **Sand Trap louvers (STL):** designed to trap sand and other debris from entering the building. They are typically made of extruded aluminum (std.) or SST or galvanized steel as an option. Their blades are Vertical, shaped to create barrier for the sand particles in the intake air, which helps to trap the debris.
- **Acoustic louvers (AAL):** stand guard against unwanted noise while managing air intake and exhaust. Their blades are stuffed with sound-absorbing material, helps to absorb sound waves & reduce their intensity.



FAL



STL



AAL

Options:

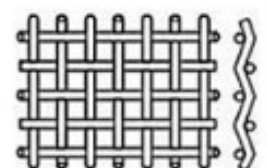
- Screens: Bird Screen or insect screen, the screen can be made of any requested material.
- Filters: The filter can be Aluminum filter of 1-inch or 2-inch thick, Bag filter, Hepa filter. The filter can be add the back of the louver or in a Unit.
- Package Unit (FAU): Air Louvers usually come in the form of a package of louvers and other options (screens, filters, dampers, sleeves,...), these units can be customized with a variety of options.
- Pent House: A penthouse is a rooftop structure featuring louvers on all four sides or maybe required as a whole structure surrounding a specific room from 2 or 3 sides or the whole building building.



Screens

Air fresh Louvers usually come accompanied with various screen options.

- Bird Screen (Std.): Anti Bird Netting wire mesh grid used behind louvers to keep out birds (13x13x0.6mm).
- Insect screen: Anti-Insect grid wire mesh used to keep insects out (3x3x0.6mm).

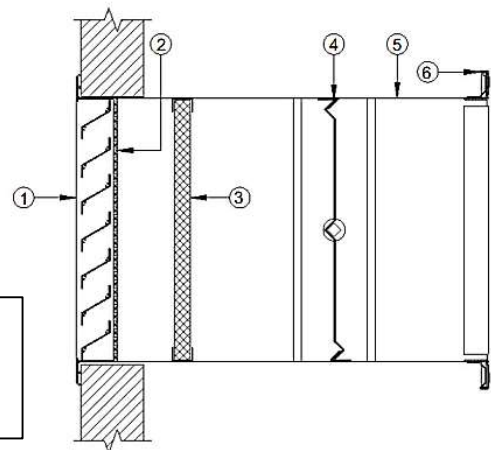


Fresh Air Unit:

- Type of air louver: The air intake louver can be a weatherproof louver, a sand trap louver, or another type of louver.
- Type of screen: Bird Screen or insect screen, the screen can be made of aluminum, galvanized steel, or another material.
- Type of filter: The filter can be Aluminum filter of 1-inch or 2-inch thick, Bag filter, Hepa filter.
- Type of Damper: The damper can be VCD, MVCD, BDD, FD or any other damper.
- Type of unit end Connection: Unit end connection can be a flat edge, a lip, or a flange for connecting to a duct.
- Method of filter installation: The filter can be installed with a door and cam or with a cam.
- Location of the door: The door can be located on the top, bottom, right, or left side of the unit.
- Location of the damper handle: The damper handle can be located on the right or left side of the unit.
- Distances in the depth direction: The distances between the components in the unit must be considered when determining the overall depth of the unit.



1. Fresh Air Louver.
2. Bird or Insect Screen (Galvanized Steel).
3. Aluminum Filter (Washable) 1 or 2 Inches.
4. Volume Control Damper (or any type of dampers)
5. GI Sleeve (20 gauge minimum).
6. Duct Mate Flange (MS Steel Angle) 35 x 30mm.



Note: All arrangements can be customized to suit the required application in terms of components, filter type, thickness & operation, damper details & operation, and sleeve details.

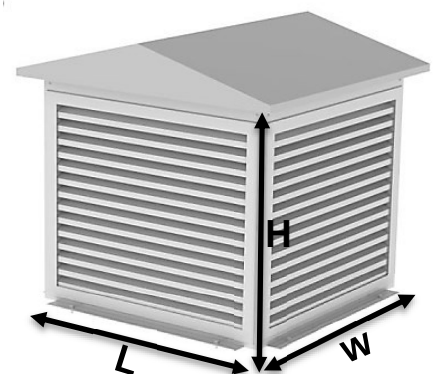
Pent House

Hood units designed for air intake, pressure-relief and exhaust applications, mounted on roofs.

Construction: heavy gauge extruded aluminum. The blades are of 1.2mm thick extruded aluminum and are polyester powder coated with your choice of color (mill finish is standard)

It Can be Customized to suit the required application including:

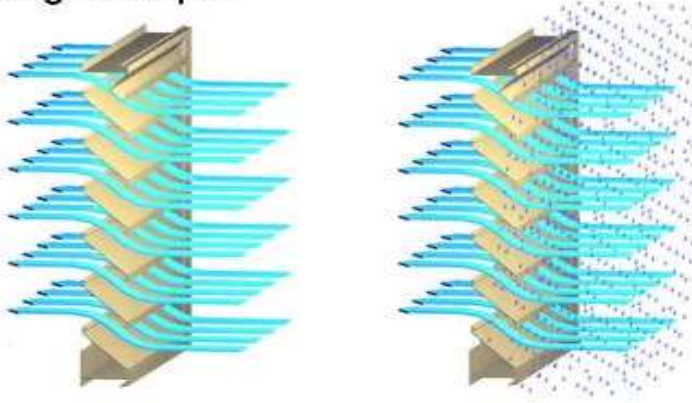
- **Type of air louver:** Can be a weatherproof louver, a sand trap louver, or another type of louver or even blank sheet.
- **No. of louvers:** Can be from 2, 3 or 4 sides.
- **Type of screen:** Bird Screen or insect screen or it can be customized.
- **Filter option:** Can be Aluminum filter of 1-inch or 2-inch thick.
- **Filter location:** Can be behind each louver or at the Bottom of PH.
- **Roof Top construction:** Can be prismatic shape (Std), flat, Triangular from 1 side or can be customized as request.



Weatherproof Louver (FAL)

Weatherproof louvers are designed to prevent rain from entering the building. They are typically made of extruded aluminum (Std.) or galvanized steel and have blades that are angled downward 30° to deflect rain this method is called Cascade method.

Working Principle



FAL-W



FAL-D

Performance

Blade spacing and inclination angle are the key factors affecting the performance of a louver. We have 3 primary models:

- **FAL-40:** Offers 40mm blade spacing and 30° Inclination.
- **FAL-55:** offers 55mm blade Spacing and 30° Inclination.
- **FAL-90:** Offers 90mm blade spacing and 45° Inclination.

Note:

We have 2 models for FAL-90, 1 suitable for wall mounting (FAL-90-W) and the other suitable for duct mounting (FAL-90-D)

Maximum Single Module size is 2.5X2.5m. Bigger sizes shall be constructed of multiple modules.

Maximum blade length for FAL-40 is 1.5m and for FAL-55 is 1.8m. For wider louvers middle Mullion (3cm wide) shall be used.

Free Area Table (FAL-40)

H	W													
mm	300	450	600	750	900	1050	1200	1350	1500	1650	1800	2000	2250	2500
300	0.033	0.050	0.067	0.084	0.101	0.118	0.135	0.152	0.169	0.182	0.199	0.222	0.250	0.279
450	0.054	0.083	0.111	0.140	0.168	0.196	0.225	0.253	0.281	0.304	0.332	0.370	0.417	0.465
600	0.076	0.116	0.156	0.195	0.235	0.275	0.314	0.354	0.394	0.426	0.465	0.518	0.584	0.651
750	0.095	0.145	0.194	0.244	0.294	0.343	0.393	0.442	0.492	0.532	0.581	0.648	0.730	0.813
900	0.114	0.174	0.233	0.293	0.353	0.412	0.472	0.531	0.591	0.638	0.698	0.777	0.877	0.976
1050	0.136	0.207	0.278	0.349	0.420	0.491	0.562	0.632	0.703	0.760	0.831	0.926	1.044	1.162
1200	0.158	0.240	0.322	0.405	0.487	0.569	0.651	0.734	0.816	0.882	0.964	1.074	1.211	1.348
1350	0.177	0.269	0.361	0.453	0.545	0.638	0.730	0.822	0.914	0.988	1.080	1.203	1.356	1.510
1500	0.196	0.298	0.400	0.502	0.604	0.706	0.809	0.911	1.013	1.095	1.197	1.333	1.503	1.673
1650	0.218	0.331	0.445	0.558	0.672	0.785	0.898	1.012	1.125	1.216	1.330	1.481	1.670	1.859
1800	0.239	0.364	0.489	0.614	0.739	0.863	0.988	1.113	1.238	1.338	1.463	1.629	1.837	2.045
2000	0.267	0.406	0.545	0.684	0.823	0.962	1.101	1.240	1.379	1.490	1.629	1.814	2.046	2.277
2250	0.299	0.455	0.611	0.767	0.923	1.079	1.235	1.391	1.547	1.672	1.828	2.036	2.296	2.556
2500	0.332	0.505	0.678	0.851	1.024	1.197	1.370	1.543	1.716	1.855	2.028	2.258	2.547	2.835

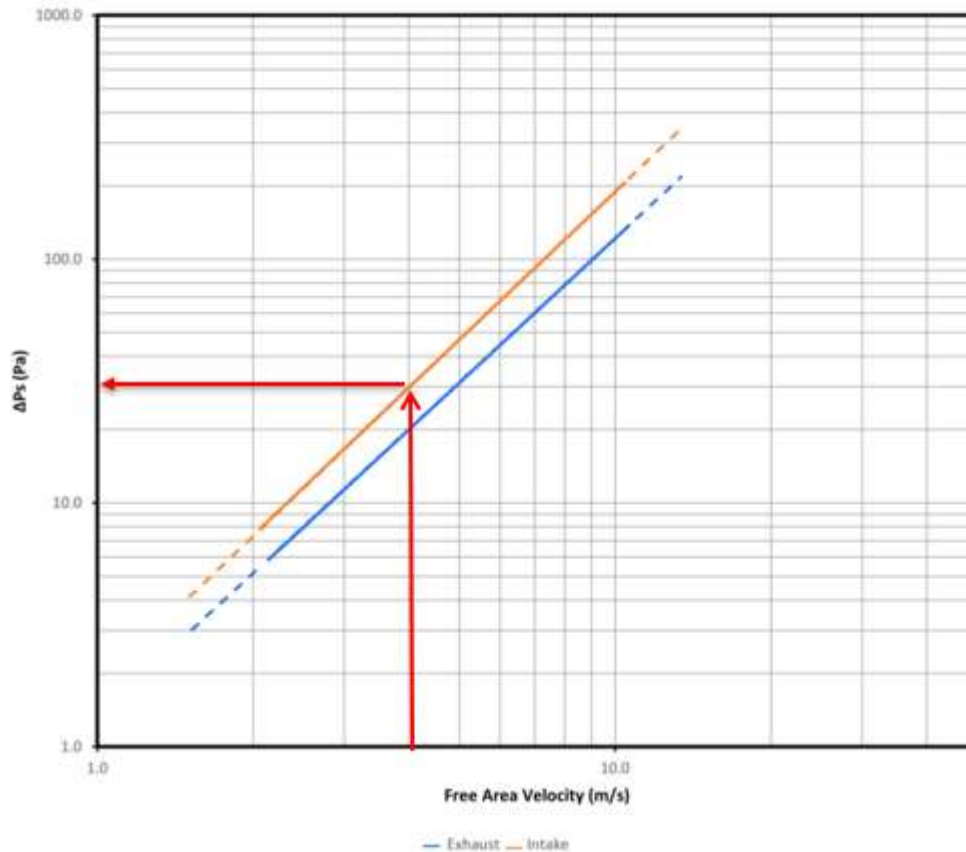
Free Area is given in m², Based on nominal neck size with 10mm clearance.

To determine the pressure:

First select the suitable Table, then locate your desired louver velocity (m/s) on the X-axis and draw a straight line across the graph. Where this line meets the specific louver Curve/line on the graph, you'll find the corresponding pressure drop value in Pa on the Y-axis.

Pressure Drop (FAL-40)

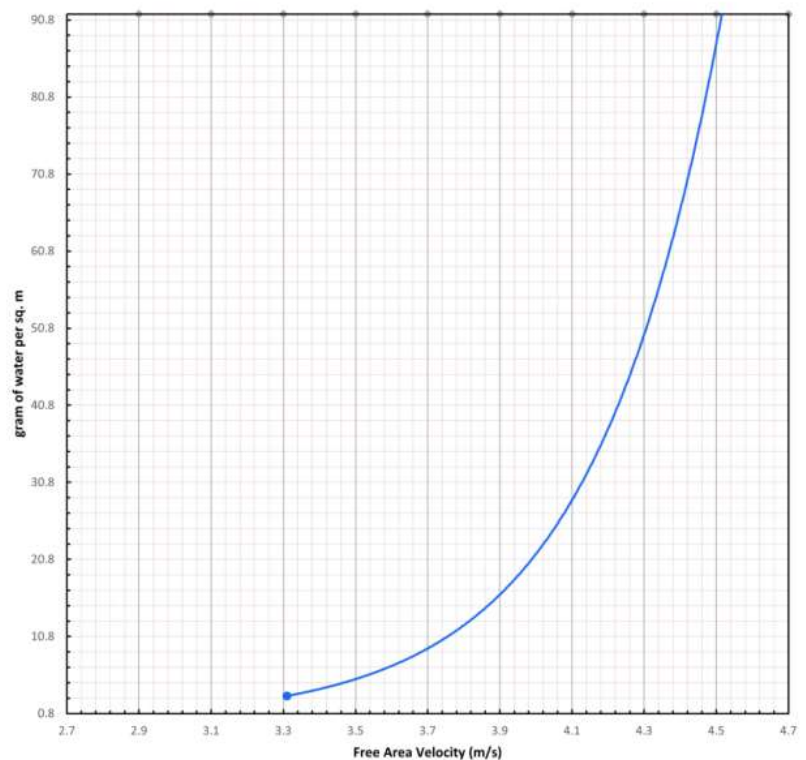
Test Sample was 1220X1220mm Outer Frame Size



You can do the same to determine the Water penetration value (g/m²).

Water Penetration per Unit Free Area (FAL-55)

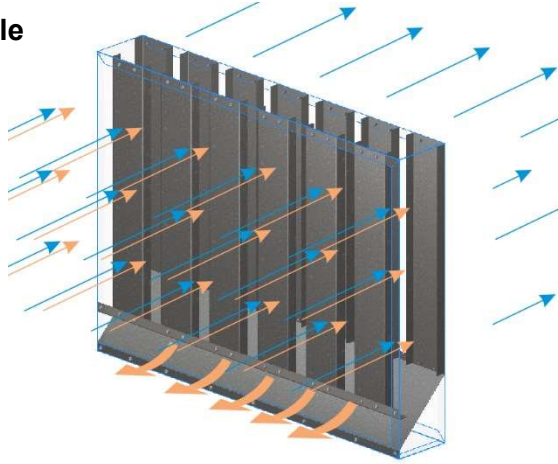
Test Sample was 1220X1220mm Outer Frame Size



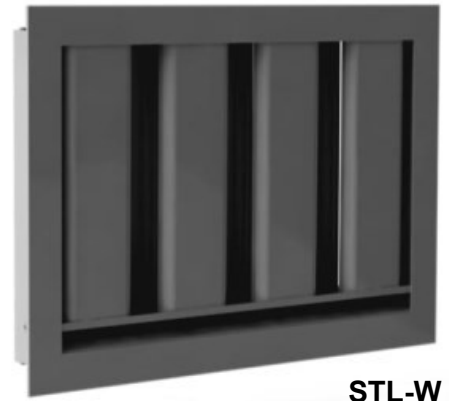
Sand Trap Louvers

Sand trap louvers are designed to trap sand and other debris from entering the building. They are typically made of extruded aluminum (Std.) or galvanized steel or SST upon request.

Working Principle



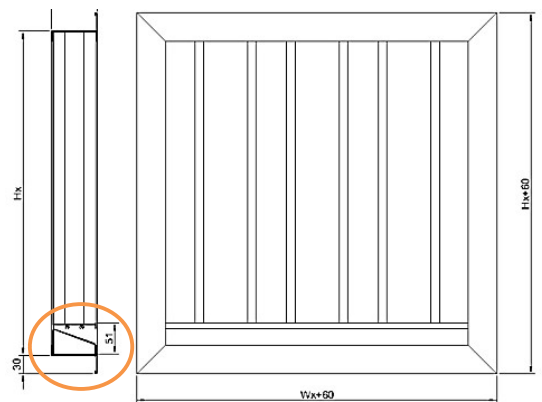
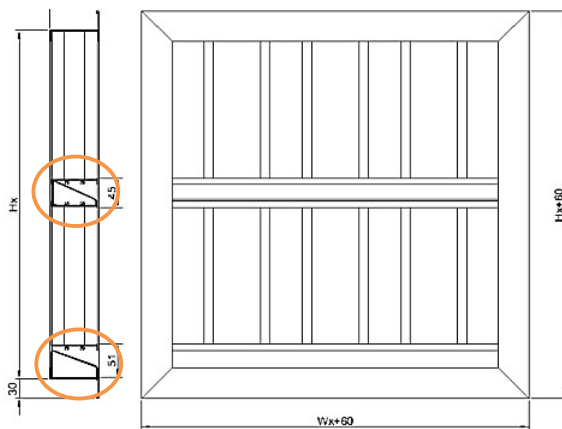
The louver utilizes 2 Sets of blades design, featuring inverted C-channel profiles mounted vertically in opposing directions. Self-cleaning holes strategically placed at the Bottom to facilitate effortless sand removal.



STL-W



STL-D



Sand trap louvers typically feature a sand chute at the bottom for efficient sand removal. Additional chute is integrated at the center of the louver for louvers exceeding 1.5 meters in height to optimize sand discharge.

For any Further clarification you scan the QR Code.

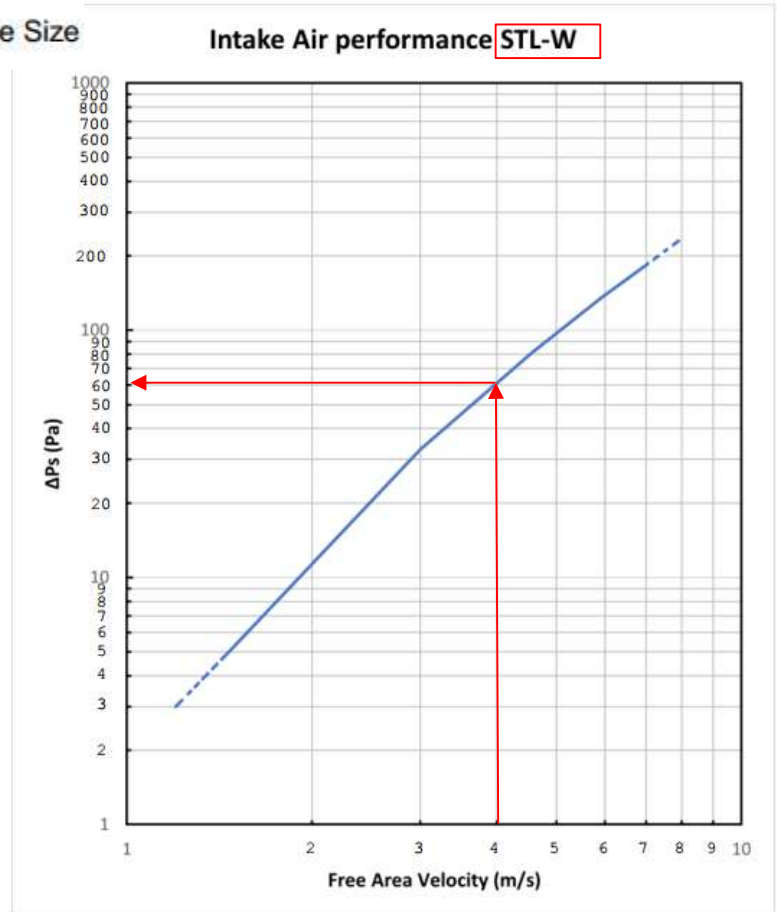


Performance

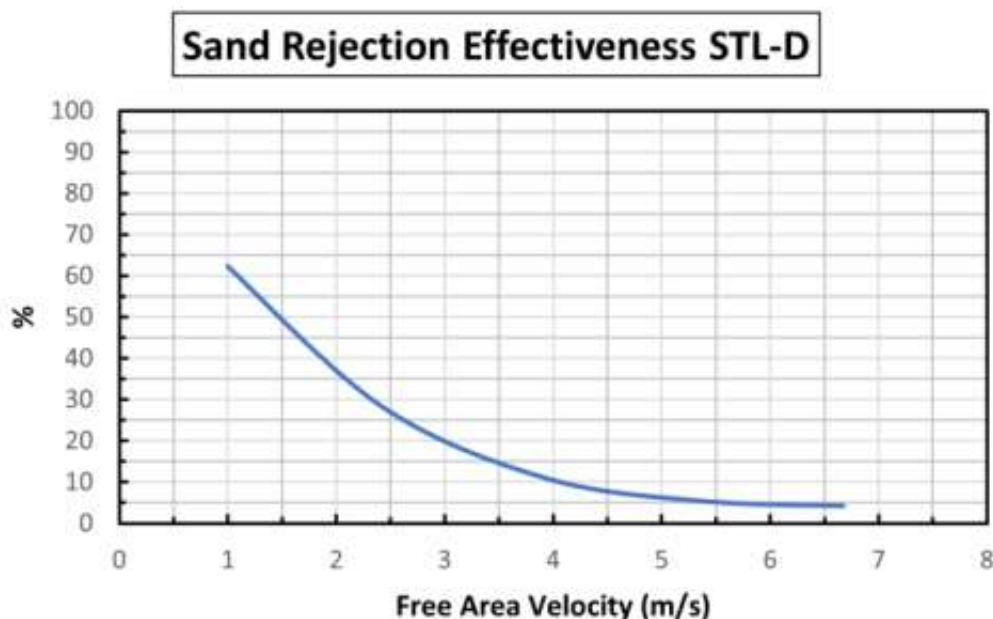
Similar to FAL, First select the suitable Table, then locate your desired louver velocity (m/s) on the X-axis and draw a straight line across the graph. Where this line meets the specific louver Curve/line on the graph, you'll find the corresponding pressure drop value in Pa on the Y-axis.

Pressure Drop (STL-W)

Test Sample was 1220X1220mm Outer Frame Size

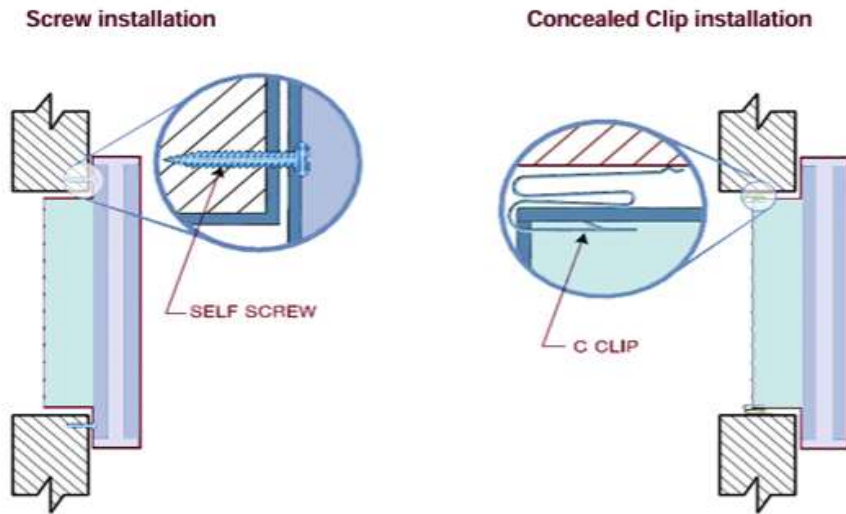


- You can do the same to determine the Sand rejection effectiveness value (%).

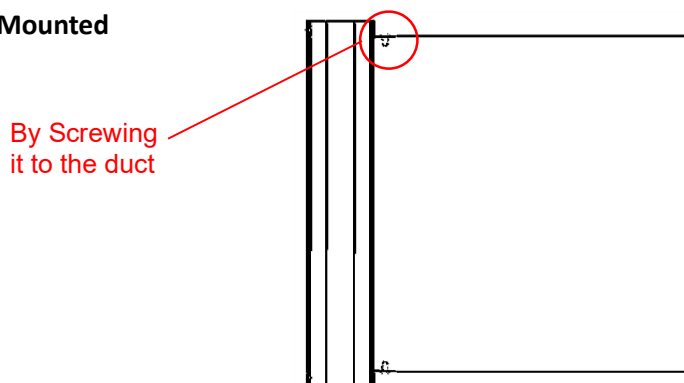


Installation

Fist: Wall Mounted



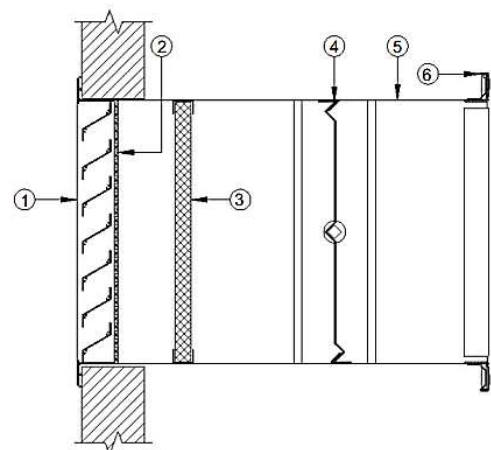
Second: Duct Mounted



Third: Fresh Air Unit:

Air fresh units are typically installed by professionals. The installation process involves the following steps:

1. Mount the air intake louver to the wall.
2. Install the damper.
3. Install the filter.
4. Connect the unit to the ductwork.



Maintenance

Air fresh units should be cleaned and maintained regularly. The filter should be cleaned or replaced as needed. The blades of the air intake louver should be inspected for damage and cleaned as needed.

To simplify unit selection, please download our comprehensive fresh air unit model guide via the QR code. Use the convenient checkboxes to pinpoint the model that best aligns with your specific requirements.

